

International Finance

Lecture II - February 24th

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IES FSV UK – Spring 2026

Moodle: <https://dl1.cuni.cz/course/view.php?id=4605>

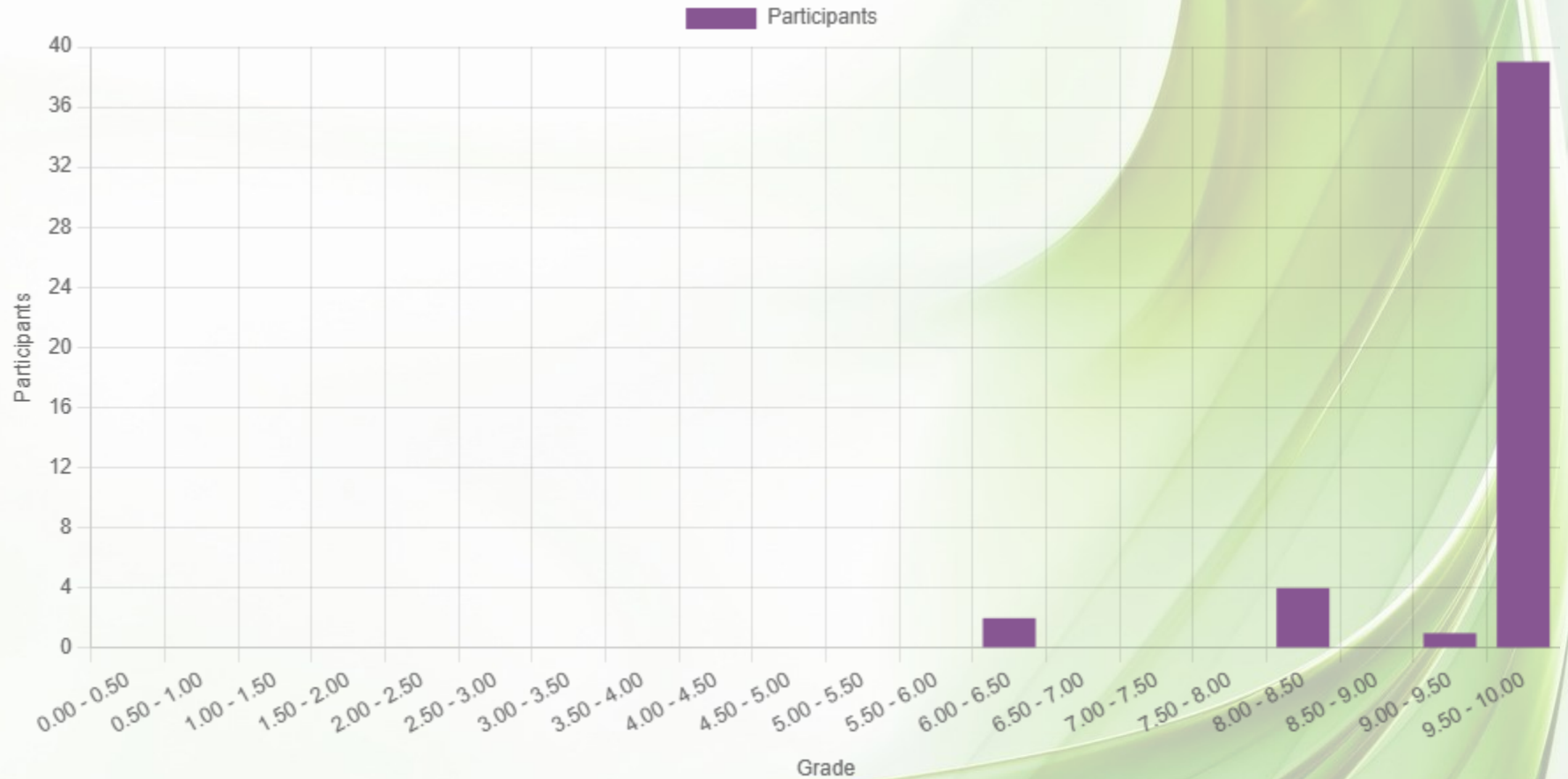
(International Finance JEB050/JPB335)

Course enrolment key: **PostUSD?2026**

Outline

- Course organization
 - Attendance, quiz
 - Quiz #2: will be open by tomorrow (Wednesday) midnight
 - The usual deadline: next Monday midnight
- A brief excursion into the history of international finance
 - From safer payments to international lending
 - Bill of exchange
- How are payments done?
 - Correspondent banking
 - SWIFT
 - TARGET
- A few remarks on trade finance
 - Factoring and forfaiting

Quiz #1



52 participants / 54 attempts

Reading for the Next Week:

- Additional materials on money, multiplication process – for those of you without prior knowledge of monetary economics
- Cipriani, Marco, Linda S. Goldberg, and Gabriele La Spada. 2023. "Financial Sanctions, SWIFT, and the Architecture of the International Payment System."
 - Can you briefly outline the difference between intra-country payments and cross-border payments? What is “correspondent banking”? What is the role of SWIFT in cross-border transactions?
 - Are there any competitors to SWIFT?
 - What is the meaning of the following acronyms: OFAC, AML, SWIFT?
 - How common are sanctions that restrict access of target countries to the infrastructures and institutions that execute international payments?

Course Outline (Changes Possible)

1. Introduction to international finance. Foreign exchange market: main actors, their role, types of interactions. Brief outline of the history of international finance. (February 17th)
2. **International transfers/payments. Channels, related instruments.** (February 24th)
3. Balance of payments structure and national income accounting for an open economy. International investment position (March 3rd)
4. Introduction to exchange rate determination: asset approach. Covered and uncovered interest rate parity. (March 10th)
5. Money, interest rates and exchange rates. Long-run aspects of exchange rate determination. Purchasing power parity. Overshooting. (March 17th)
6. Prices and exchange rates continued: Fischer effect. Exchange rates and competitiveness: real (effective) exchange rates, Balassa-Samuelson effect. Application on the nominal and real convergence in the EU (March 24th)
7. Introduction to the AA-DD model. Temporary and permanent changes in fiscal and monetary policies. (March 31st)
8. The AA-DD model: XX curve. Application on policy analysis. (April 7th)
9. Policy options for reaching internal and external balance. Swan diagram. Fixed exchange rates: macroeconomic implications. Interventions and sterilisation. Policy trilemma. (April 14th)
10. **Monetary integration: costs and benefits. Monetary integration: effects – empirical research.** (April 21st)
11. A brief introduction to the models of balance of payments crises. (April 28th)
12. Debts and (developing) countries. **Deadline for the team assignment/paper** (May 5th)
13. The future of IMS, role of the USD. Discussion of current topics. (May 12th)
14. **Reserved for a make-up class (possibly for April 21st). (May 19th)**

Global Finance: History

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Origins of International Finance

- Two factors led to the emergence of international finance
 - Long-distance trade
 - Long-lived (productive) assets
- Markets for goods/services and markets for assets require
 - **Conversion** (if economic agents choose or are forced to use different currencies)
 - **Finance to bridge time** between an agreement and delivery, or time of production,
 - Plus: how about **diversification**?
- Plus: existence of instruments created to address the issues provides new opportunities too...

Bigger Picture: What is Needed?

- Efficient international trade/asset trading requires:
 - Safe **channels** for payments
 - Distance implies: need for safe **matching of payments / ownership transfer**
 - Trade finance: letters of credit, documentary collections
 - Time involved implies questions: can I get paid faster or restructure my exposure when needed?
 - Again, trade finance: factoring, forfaiting
 - Presence of markets/brokers for trading of the instruments used?
- Certainty over solvency/control over rights to use/transfer assets
 - Centralised registrar
 - Physical evidence (tokens, tally sticks etc.)
 - Decentralized solutions?
- **Common rules and enforceability!**

Liabilities Records: Exchequer Tallies



- In 12th - 19th centuries, the Exchequer of England used tally sticks made of wood to account for taxes owned and collected
- The stick would be split in half: the lender received 'the stock'
 - This is where the modern meaning of the word for a finance certificate originates from,
- and the debtor the other half, called the 'foil'.
- In these sticks, which are only stocks, notches show the amount deposited, the name of the lender, the county, the time of year when the money can be repaid, and the year.
- The stick's other half contained the same information. When the debt was paid, matching the two halves showed that they 'tallied, the wood's unique features and split ensuring high security against forgeries.

Examples of Instruments/Assets in History

- Of course: bullion (gold, silver)
- **Bills of exchange**
- Shares:
 - Dutch VOC
- Bonds (sovereign bonds)
-

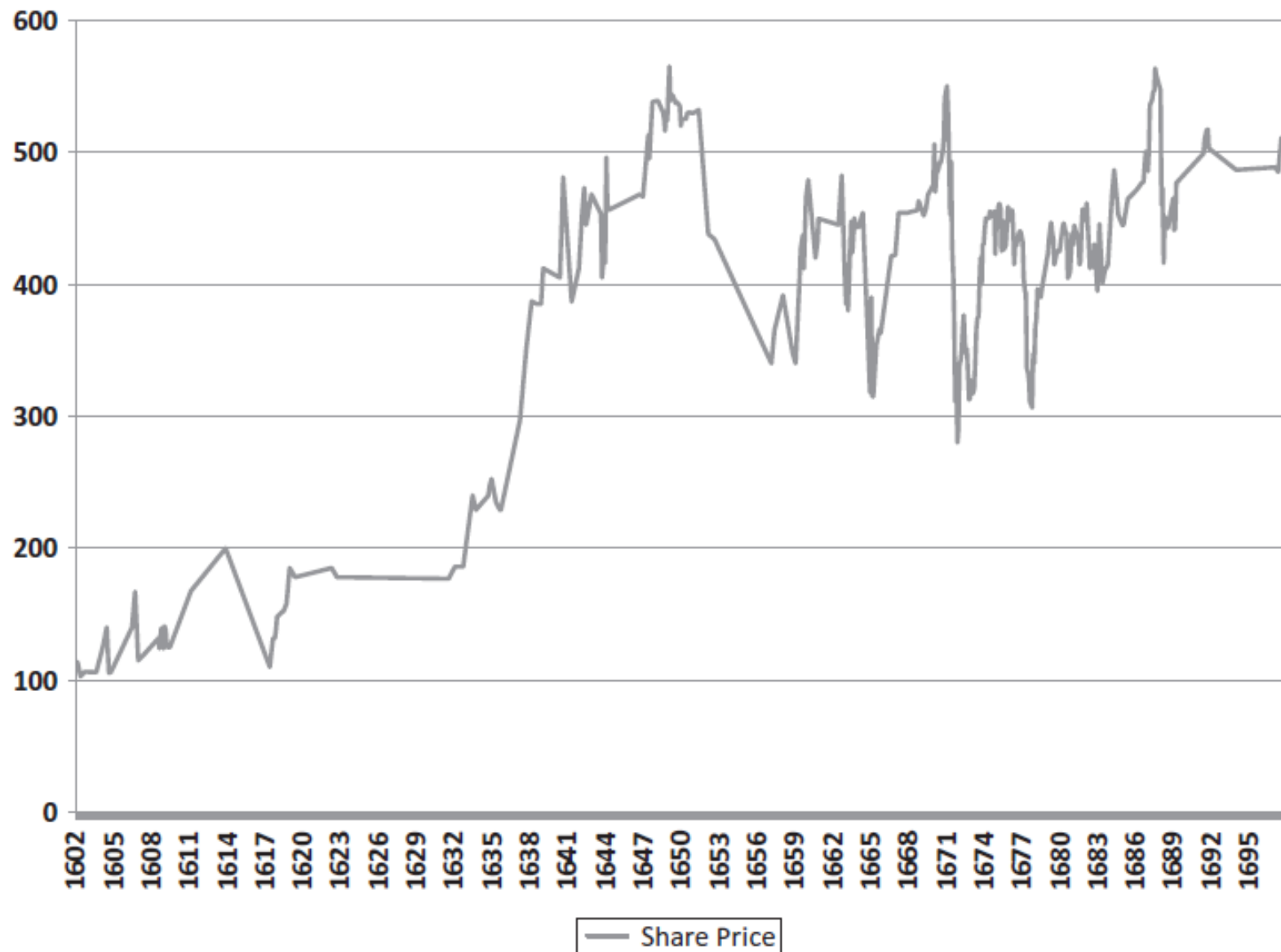


Figure 4.1 Share prices for VOC, 1602–1698 (from Petram 2011)

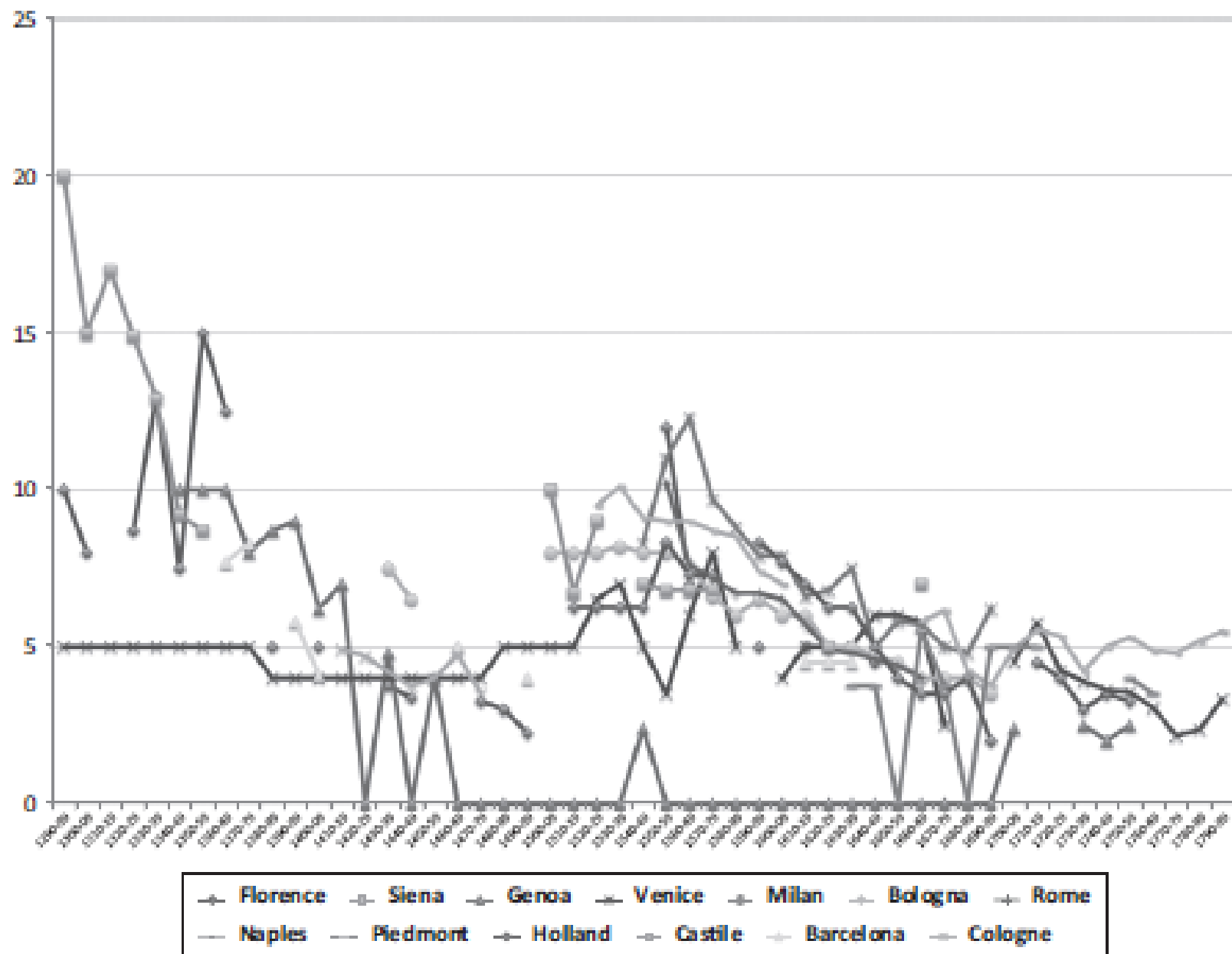


Figure 3.3 Sovereign bond yields in western Europe, 1290–1780 (from Pezzolo 2014, Figure 10.1, p. 300)

NEW APPROACHES TO ECONOMIC AND SOCIAL HISTORY

A CONCISE HISTORY OF INTERNATIONAL FINANCE

From Babylon to Bernanke

Larry Neal



Pre-History of International Finance

- At least as old as the history of international trade, i.e. several millenia!
- Example: records from ancient Sumer
- Van der Mieroop (1992)
 - Clay tablets with records about one businessman, Dumuzi-gamil who flourished **by financing long distance trade and local agriculture**
 - Some of the contracts arranged **partnerships among as many as thirty other individuals** in financing substantial trade with other ports on the Persian Gulf.
 - This continued until the **king, Rim-Sin, issued an edict declaring all debts null and void in 1788 BC**
 - Indeed: **1788 BC seems to be the first written record of a financial crisis**
 - By the way: the record shows that Dumuzi-gamil's business was ruined, all the subsequent documents concern only continuing lawsuits....

Source: Neal (2015), p. 16

Birth of Modern Finance

- Neal (2015):
 - *All Italian scholars insist that the origins of modern international finance arose among the Italian city-states*
 - *Italian merchants perfected the foreign bill of exchange to finance trade throughout medieval Europe into the North and Baltic Seas, as well as with the ancient ports of the Mediterranean and Black Seas.*
 - *Their home cities also developed **the precursors of sovereign debt** in the form of perpetual annuities.*
- *Florence, Venice, Genoa*
- What was one of the main obstacles that they had to deal with?

International Finance and Crusades?

- Some historians emphasize rather the role of Popes (specifically Pope Gregory VII, 1073-1085), and **Papal finance**
- Again Neal (2015)
 - *Gregory's successors mobilize Western European believers to undertake the large-scale efforts of the Crusades into the Near East over the next two centuries, they also **followed his lead in centralizing collection of revenues from all of western Christendom and remitting them regularly to the papal Curia in Rome.***
 - *Key to the success of both papal initiatives of expansion and centralization was **mobilizing international finance, exchanging currencies collected in one part of Europe into means of payment acceptable in another.***

Remittances

- Problem: export of gold or silver often prohibited
 - You can transfer value by relying on wool, wine, etc.
- An interesting solution: Foreign Bill of Exchange

(Foreign) Bill of Exchange

- A short-term negotiable instrument
- A signed, unconditional, written order binding one party to pay a fixed sum of money to another party on demand or at a predetermined date.
- Bills of exchange are similar to checks and promissory notes—they can be drawn by individuals or banks and are generally transferable by endorsements.
- They are used in international trade to help importers and exporters fulfill transactions.
- A bill of exchange is not a contract itself
- Foreign bill of Exchange = a bill of exchange drawn in one state or country and payable in another, as one arising from foreign trade operations

Bill of Exchange from the Collections of the British Museum



Bill of Exchange: Parties

- Drawee
 - The party that pays the sum specified by the bill of exchange.
 - Usually a bank or **buyer of goods and services**.
- Payee
 - is the one who receives that sum.
- Drawer
 - is the party that obliges the drawee to pay the payee.
 - Typically account holder or **seller**.
- The drawer and the payee are the same entity unless the drawer transfers the bill of exchange to a third-party payee.

How did woolen manufacturer in Florence pay for the Supply of English wool?

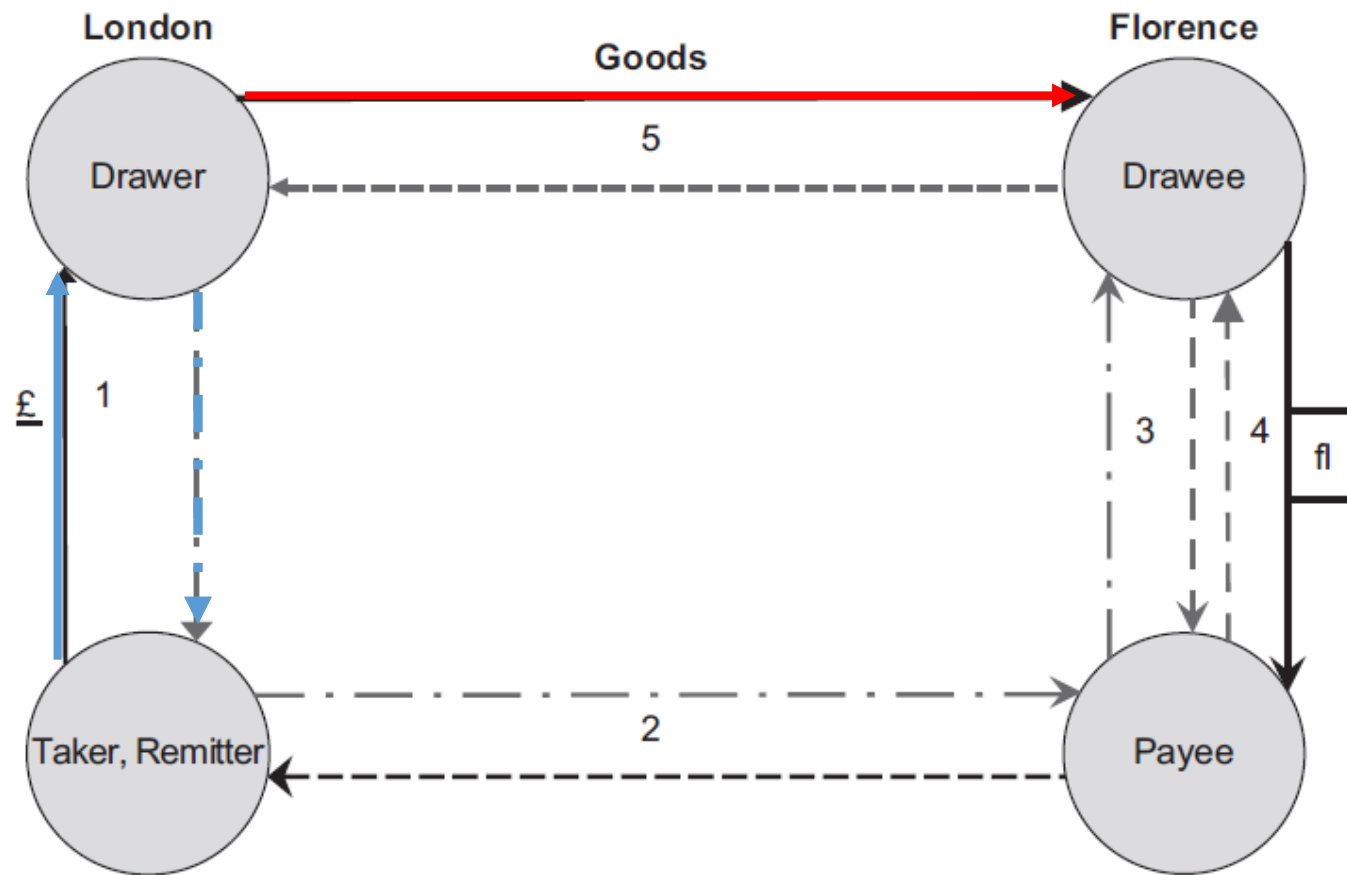


Figure 3.1 The bill of exchange as an instrument to finance trade

Trade Finance (Neal)

- Components of the transaction:
 - Italian banking firm could collect the papal dues in England and purchase a bill of Exchange from a wool merchant in London
 - The wool merchant would send the equivalent quantity of wool to Florence.
 - The weaving firm (in Florence) would pay for it in Florentine ducats to redeem the bill presented to them by the bank's representative in Florence.
- Results:
 - At the end of the transaction, the weaver in Florence paid for the English wool in ducats, extinguished the bill, and then sent it back to the wool merchant in London.
 - Meanwhile, the banking firm in Florence credited the account of its partner (or factor) in London for the payment.

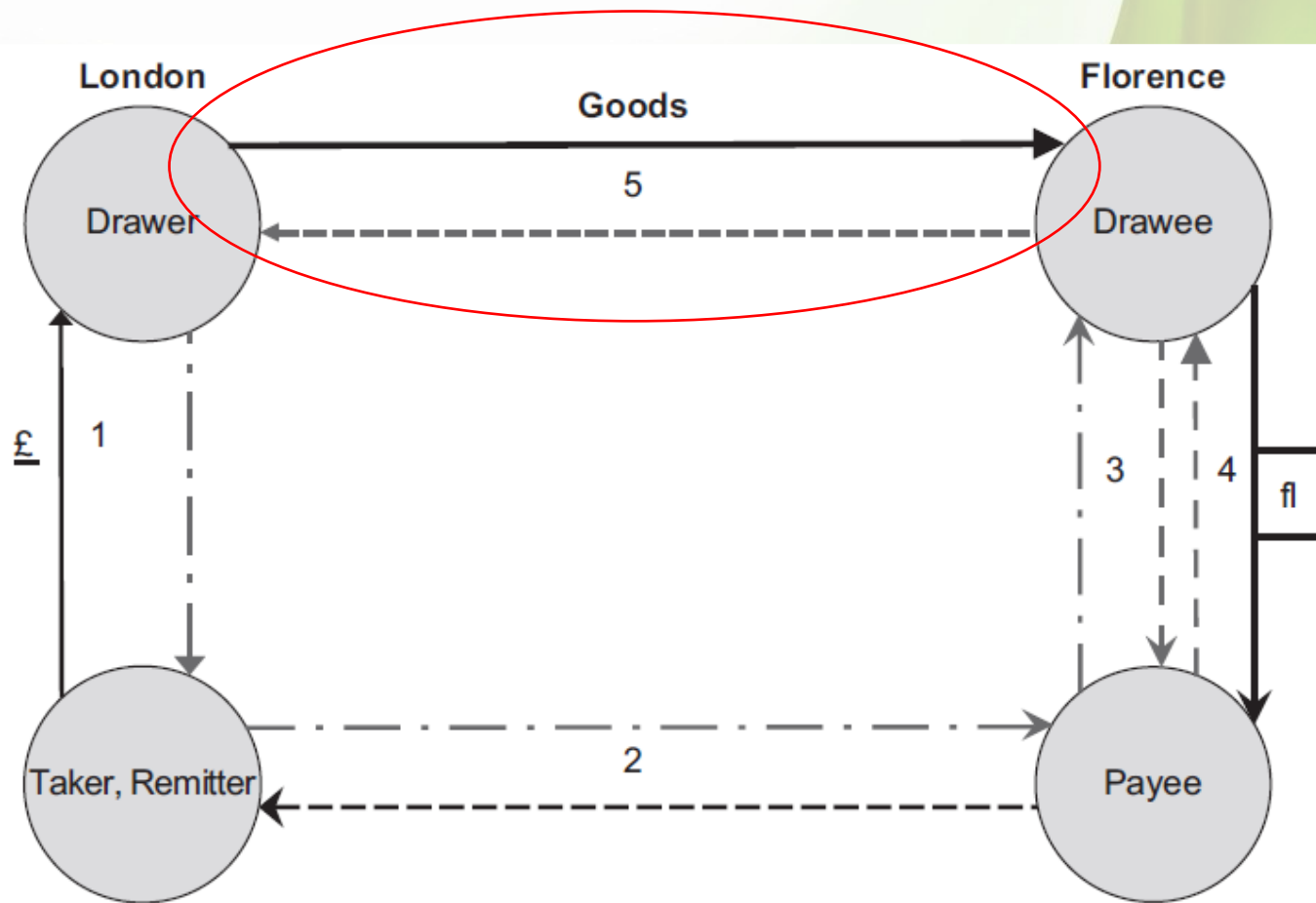


Figure 3.1 The bill of exchange as an instrument to finance trade

Bill of Exchange: As a Credit Instrument?

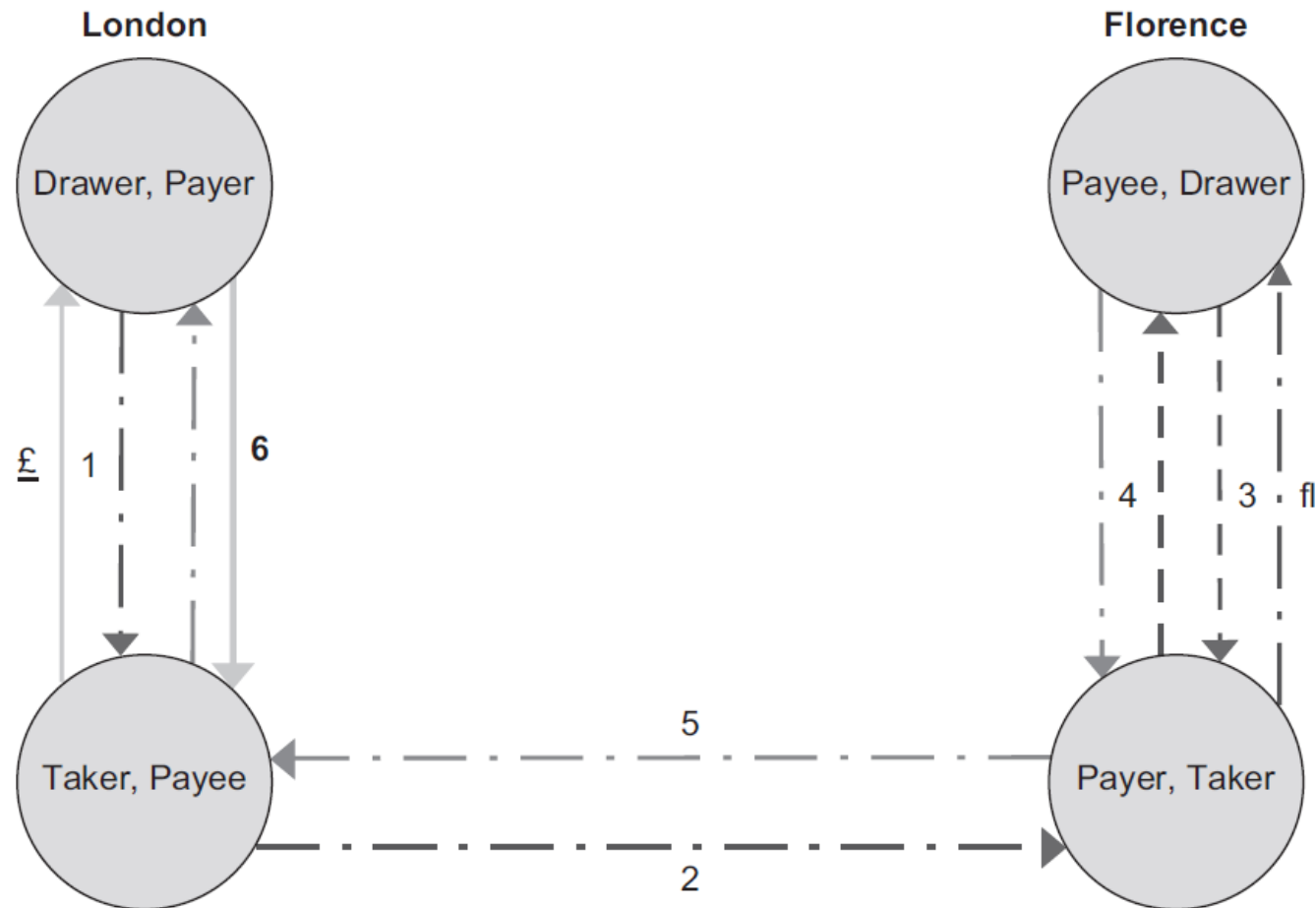


Figure 3.2 The bill of exchange as a credit instrument

So What Happens in this Case?

- **Step 1** reprises the roles of the merchant and the financier with the financier paying the merchant in pounds sterling for the bill he has drawn;
- **Step 2** the financier sends the bill to his correspondent in the foreign city, who accepts it, but holds it until it is due for payment in Florence.
 - Rather than make payment in ducats, however, he writes a “rechange” bill demanding payment of the stated amount of ducats in pounds sterling back in London.
 - If the exchange rate of pounds/ducats is the same in both cities, then the borrower in London would just be asked to repay the same amount of pounds he borrowed originally.
 - But, due to the normal difference in exchange rates between the two cities, depending which city is quoting the rate, the amount of pounds demanded in London by the “rechange” bill coming back from Florence will typically be more than the amount originally paid

“Dry Exchange” and Credit

- “dry exchange”:
 - The financier becomes the “taker” of the bill by lending a sum to the drawer, which he expects to have repaid in due course
 - The circuit of obligations is now completed by a re-exchange of a new, reciprocal bill rather than by the delivery of goods.
 - At the end of the two circuits of payment by bills, the original receiver of cash in London must pay back a specified sum to the original taker.

Summary and Conclusion

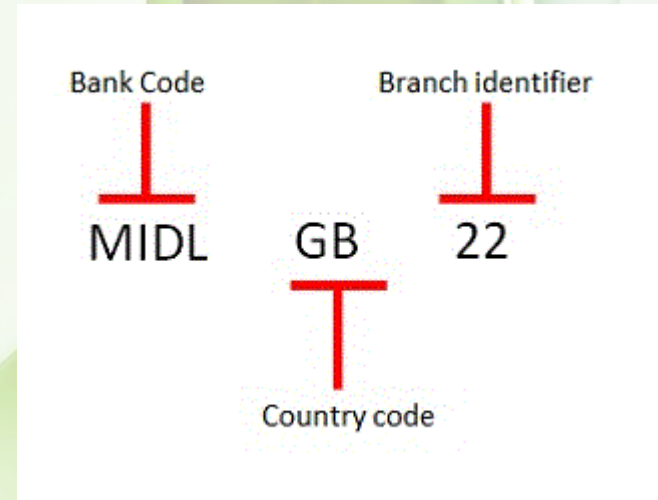
- Many modern instruments of finance (and logic behind them) have rather long history
- International financial transactions dwarf the international trade flow
 - This is not surprising – assets are involved
- In spite of all changes in global economy, the USD remains the top currency – in reported transactions as well as in forex reserves
 - The role of Chinese RMB is increasing relatively fast
 - The US reliance on financial sanctions can be seen as a threat to the willingness to use dollars
 - The relative total role of cryptocurrencies seems to be small
- In spite of globalization and computerization, forex transactions seem to be still unequally geographically distributed.

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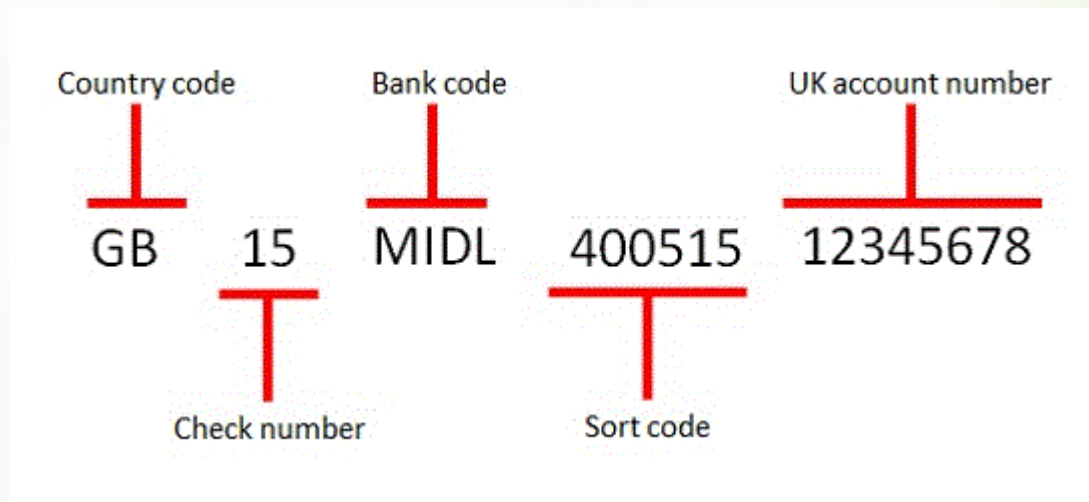
How are International Transfers
(Payments) done?

What we see as customers:

- MIDLGB22XXX
- FIOBCZPPXXX
- EQBKCZPP
- **BIC** – Bank Identifier Code



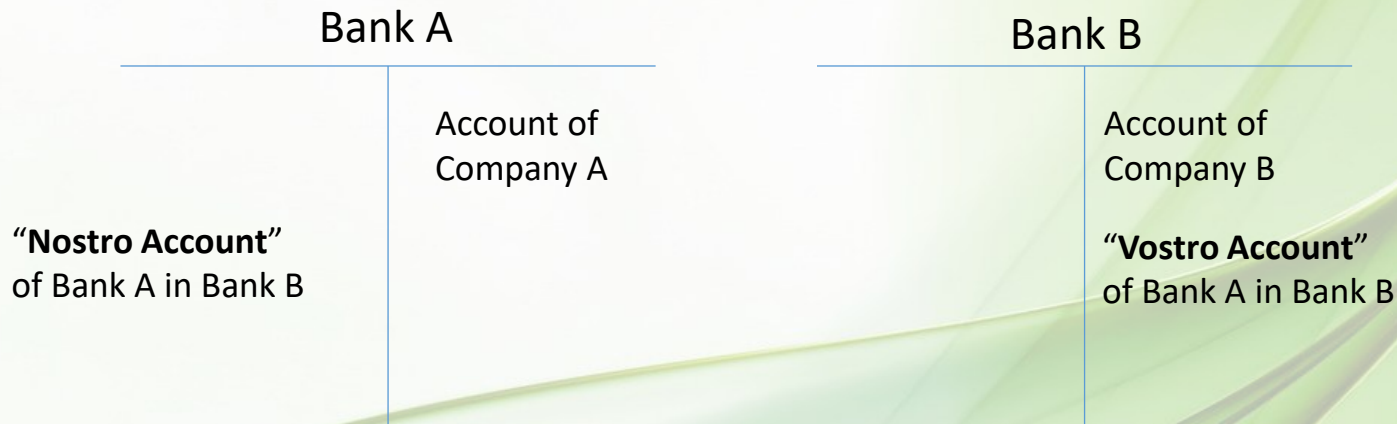
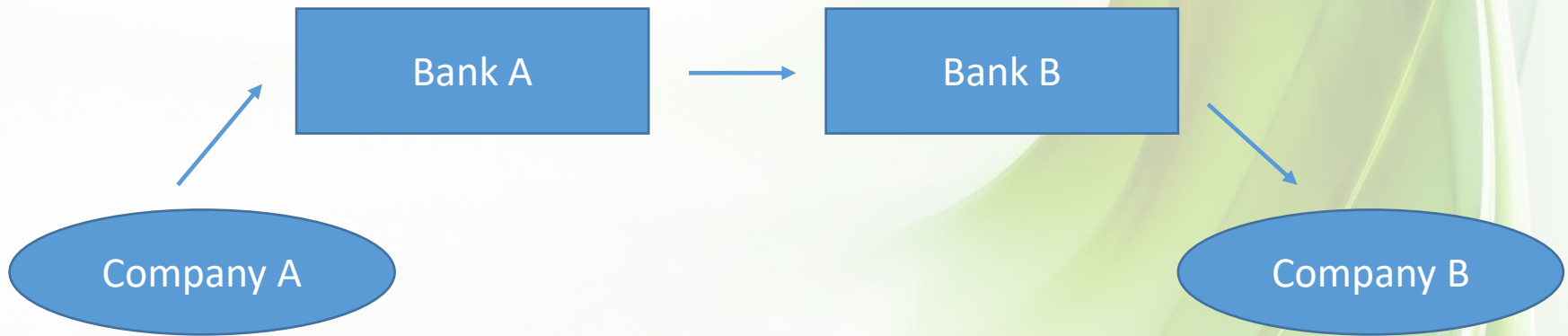
- **IBAN** - International Bank Account Number



Payment Systems

- Large-value payment systems
 - Real time, large volumes
 - Payments sent by the ordering financial institution are immediately credited to the beneficiary bank and are final and irrevocable.
 - Transactions are settled on an individual order (transaction-by-transaction) basis without netting debits with credits (gross settlement).
- Clearing systems
 - Retail payments, cost-efficient
 - Consumers or companies initiate such retail payments, and banks send files (compilations of payments) to the clearing house, which collects the transactions and, at particular cutoff times, compiles each bank's net settlement position
- Correspondent banking arrangements
 - Bilateral links between payment institutions.
 - In the euro area, correspondent banking relationships are increasingly losing ground to centralized euro settlement systems.

How International Payments are done?



All Banks to not have to have direct contacts: Loro Account

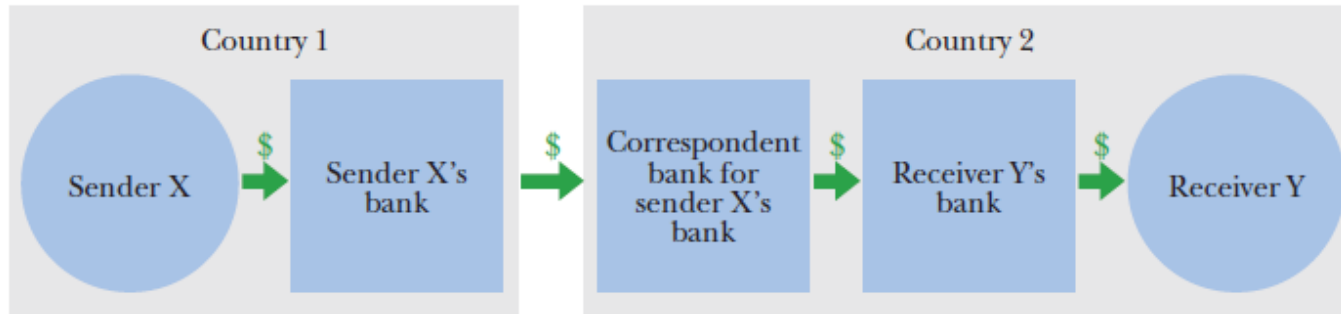
- Most correspondent banking arrangements involve at least one additional party: '**correspondent**' or 'service-providing' bank.
 - The **correspondent bank** receives payment instructions from one or both institutions and processes them via book-to-book transfers across banks.
- This involves the use of nostro, vostro and loro accounts.

Nostro account	Vostro account	Loro account
• <i>Our money with you</i> • Foreign currency account held by a domestic bank in a foreign bank	• <i>Your money with us</i> • Domestic currency account held by foreign banks in a domestic bank	• <i>Their money with them</i> • An account used by a third party bank to settle foreign exchange transactions via a domestic bank which holds a nostro account

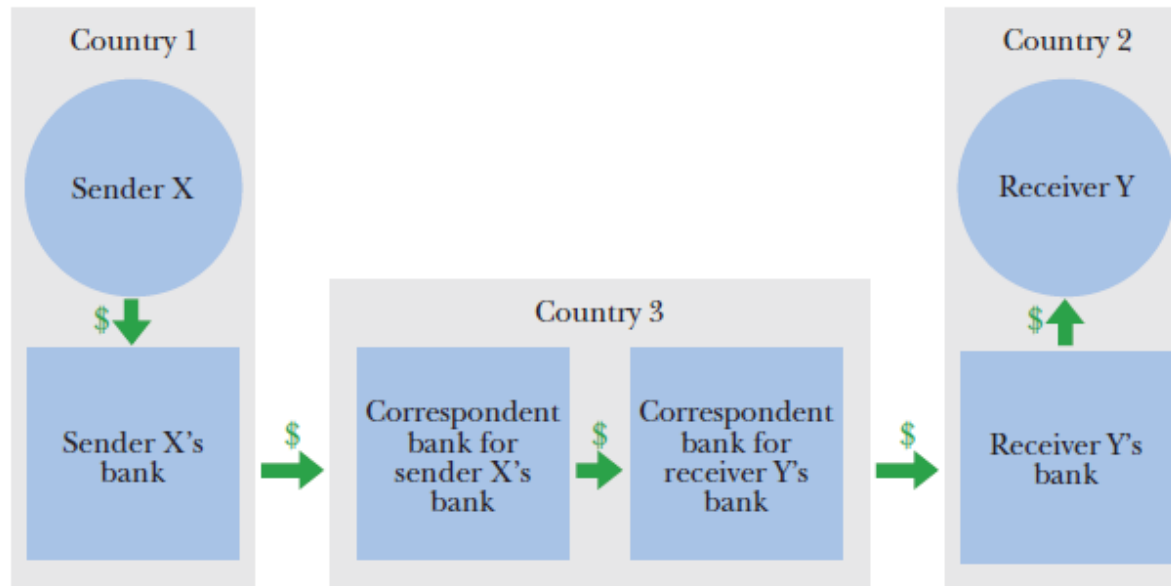
Source: ECB.

Cross-border Payments and Correspondent Banking

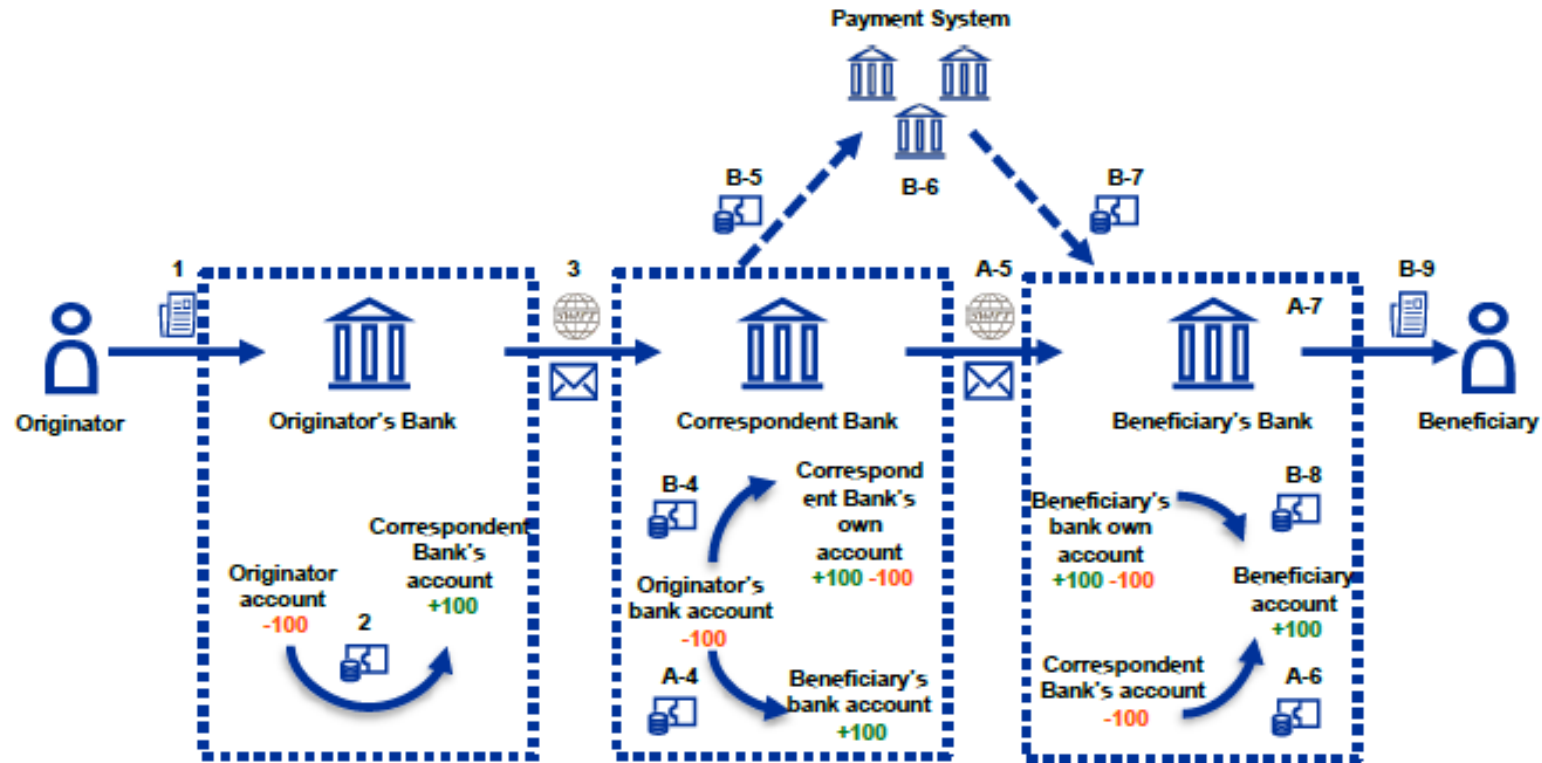
Panel A. Two countries



Panel B. Three countries



Payment Flow in Correspondent Banking



1. The originator instructs the payment
2. The originator's bank debits the account of the originator and credits the correspondent bank account
3. The originator's bank sends a SWIFT message to the correspondent bank

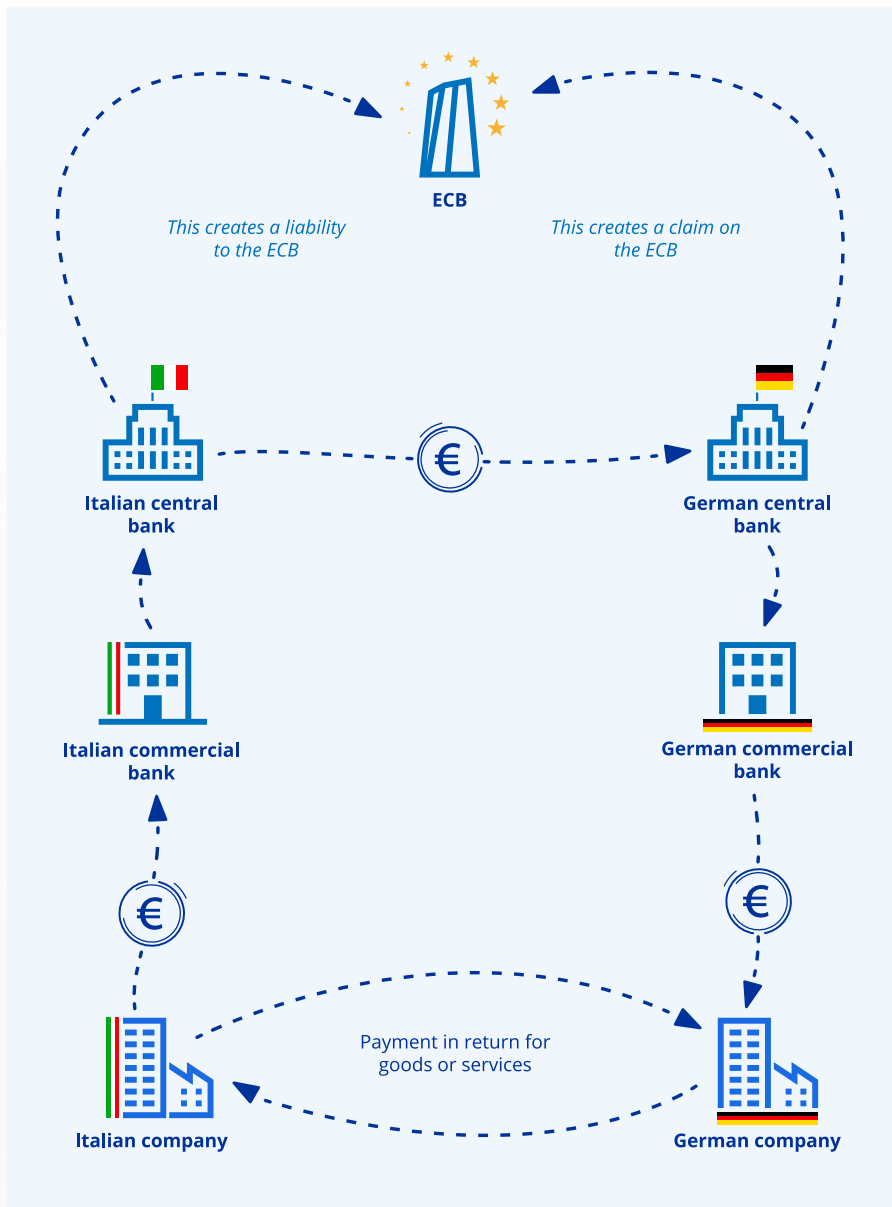
A: Use of correspondent bank only

- A-4 The correspondent bank debits the originator's bank account and credits the beneficiary's bank accounts
- A-5 The correspondent bank sends a SWIFT message to the beneficiary's bank
- A-6 The beneficiary's bank debits the correspondent bank's account and credits the beneficiary's account
- A-7 The beneficiary is informed that the transaction is completed

B: Involvement of payment system

- B-4 The correspondent bank debits the originator's account and credits its own account
- B-5 The correspondent bank sends a payment instruction to the payment system
- B-6 The payment is cleared and settled via the payment system
- B-7 The beneficiary's bank receives the funds on its own account
- B-8 The beneficiary's bank credits the beneficiary's account
- B-9 The beneficiary is informed that the transaction is completed

Large-value Payment Systems: TARGET2



TARGET2 ("Trans-European Automated Real-time Gross settlement Express Transfer system 2") is a system that moves money from one bank to another, both within countries and across borders.

Central banks and commercial banks use it to process payments in euro and move money safely and easily between them.

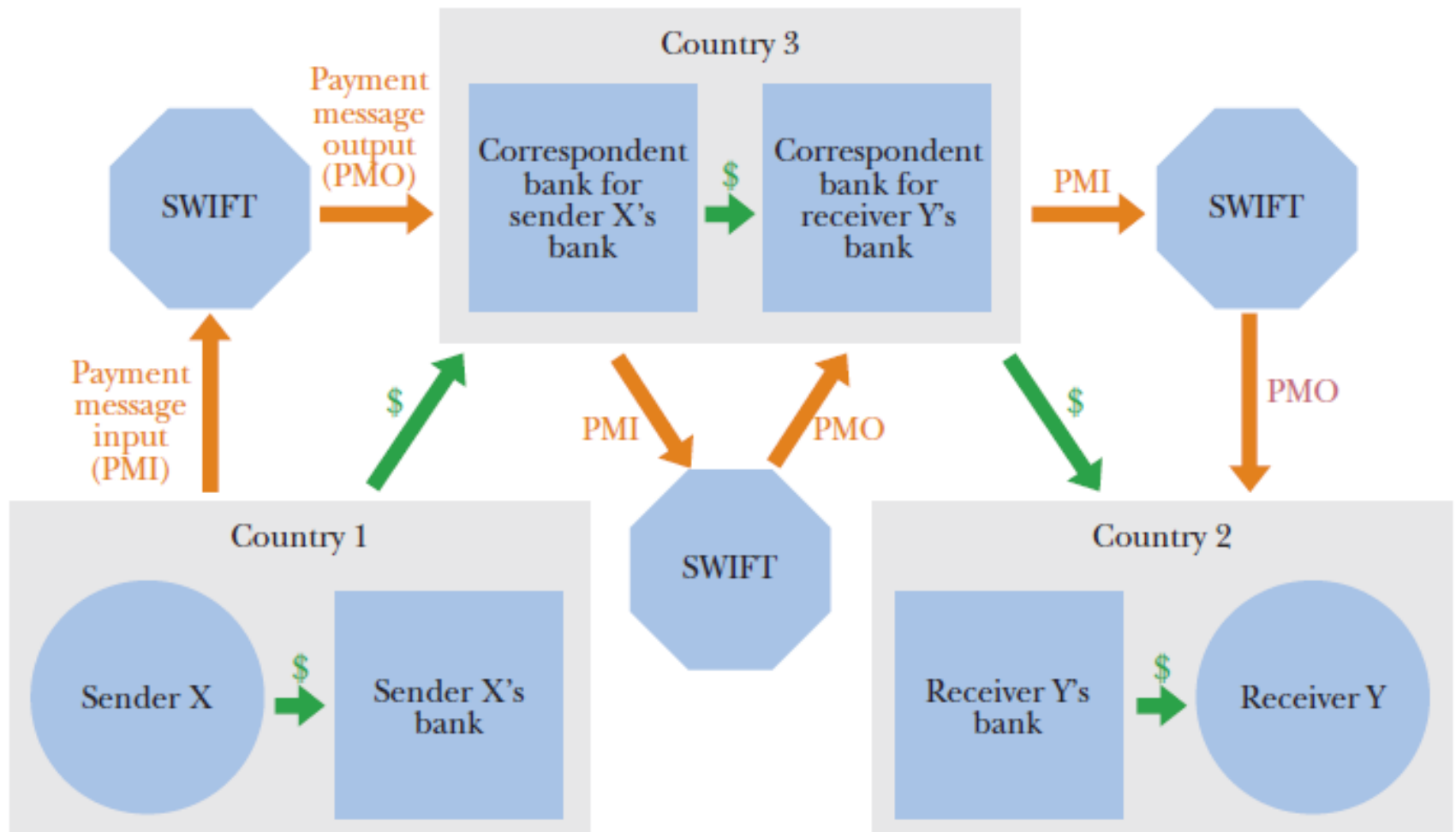
A single technical platform, shared by all participating central banks.



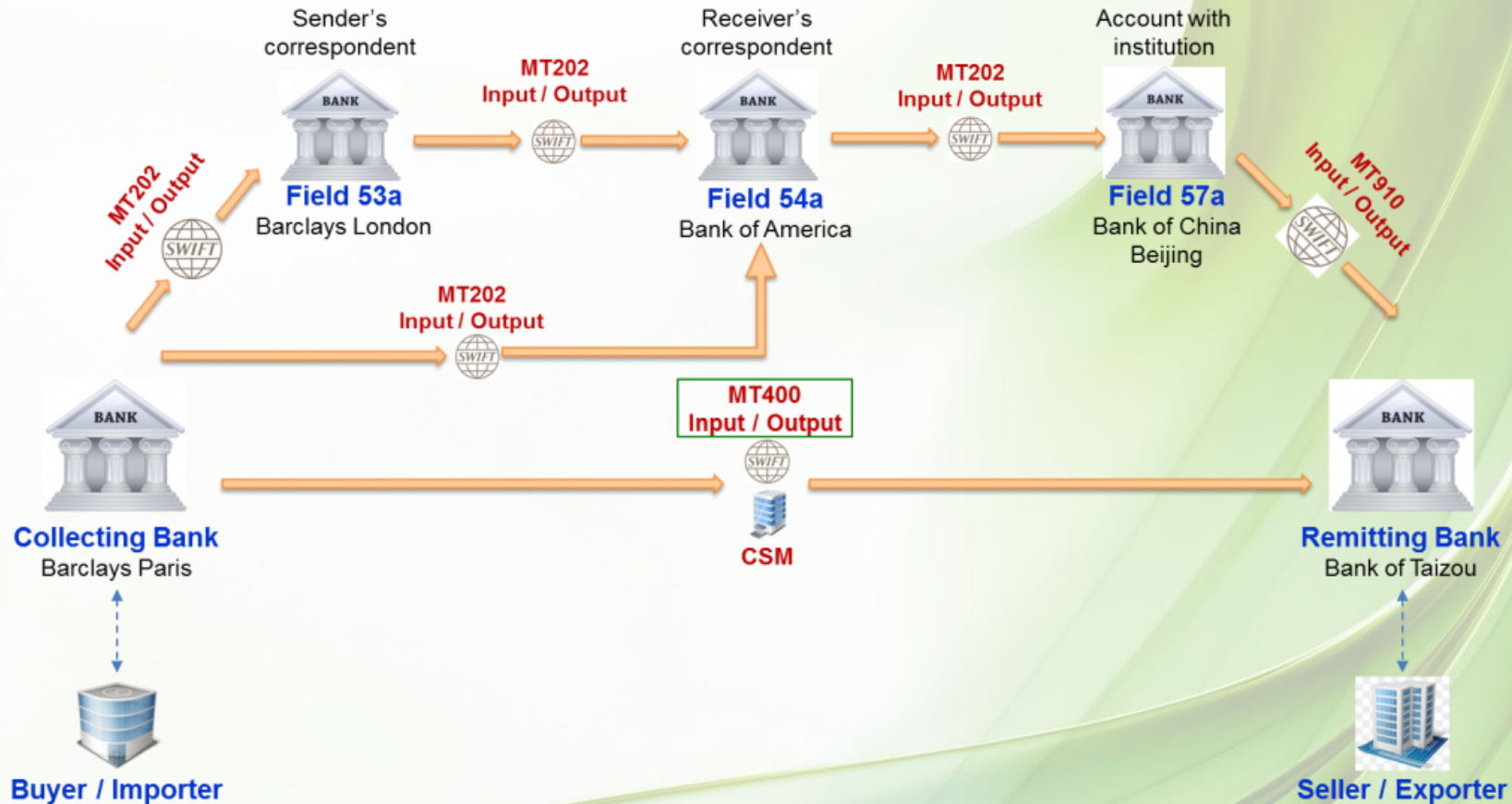
SWIFT

- the **Society for Worldwide Interbank Financial Telecommunication**
 - Founded in Brussels, 1973 (banks from 15 countries initially)
 - a member-owned cooperative (Belgian law) through which the financial world conducts its business operations with speed, certainty and confidence.
 - **Secure messaging network**
 - “the most secure trusted third-party network in the world”
 - More than 11 thousands banking organisations, securities institutions and corporate customers in 200+ countries every day
 - SWIFT is overseen by the G-10 central banks (Belgium, Canada, France, Germany, Italy, Japan, The Netherlands, United Kingdom, United States, Switzerland, and Sweden), as well as the European Central Bank, with its lead overseer being the National Bank of Belgium.

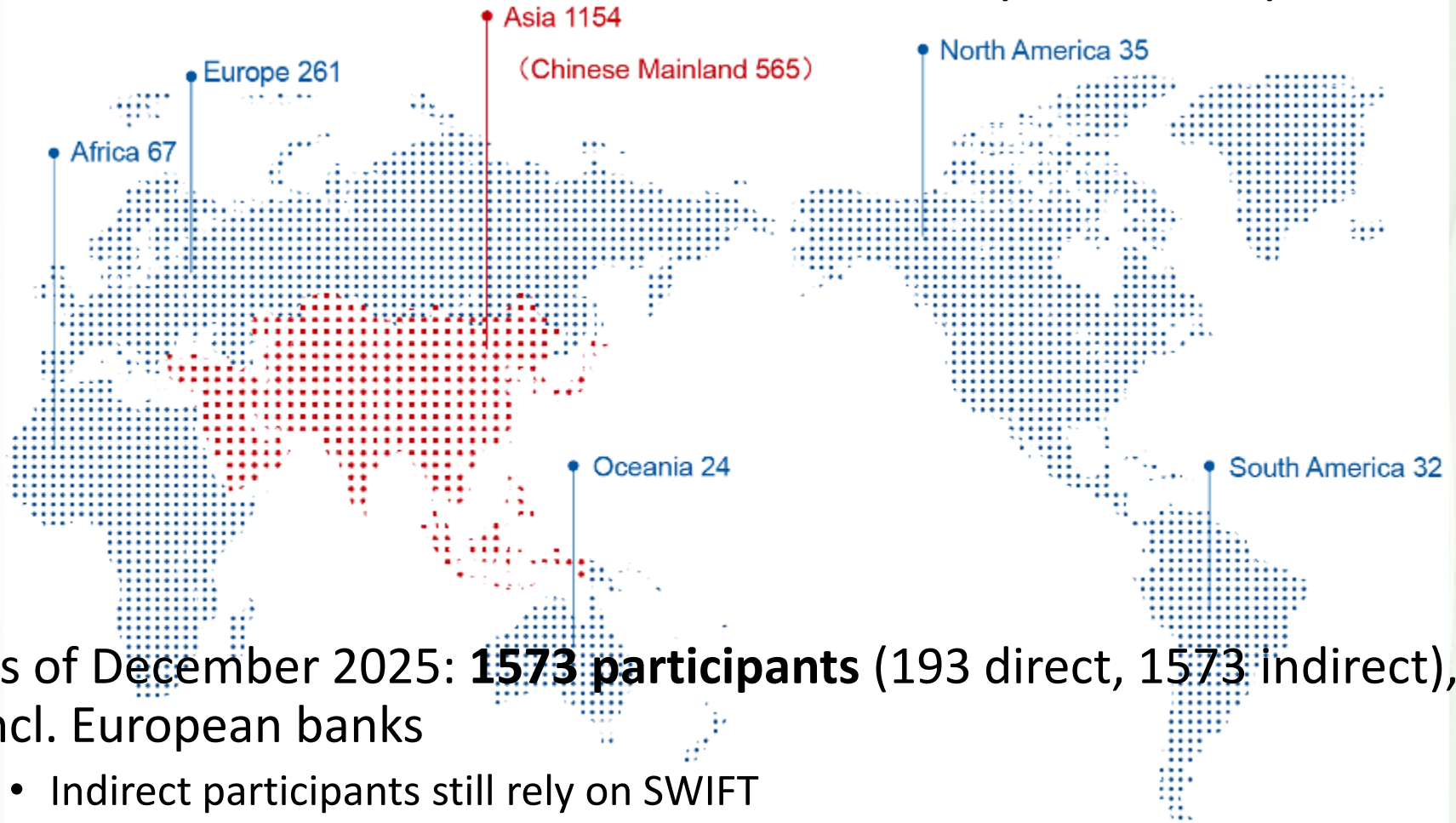
The Role of SWIFT in Cross-border Payments



Example: A Company from China exports goods or services to a French company



CIPS: Cross-Border Interbank Payment System



- As of December 2025: **1573 participants** (193 direct, 1573 indirect), incl. European banks
 - Indirect participants still rely on SWIFT
- CIPS still quite small relative to SWIFT
 - 2025: Total volume of transactions at RMB 180 trillion
 - SWIFT: about USD 7.5 trillion daily (net, 2022)
- Web: <https://www.cips.com.cn>

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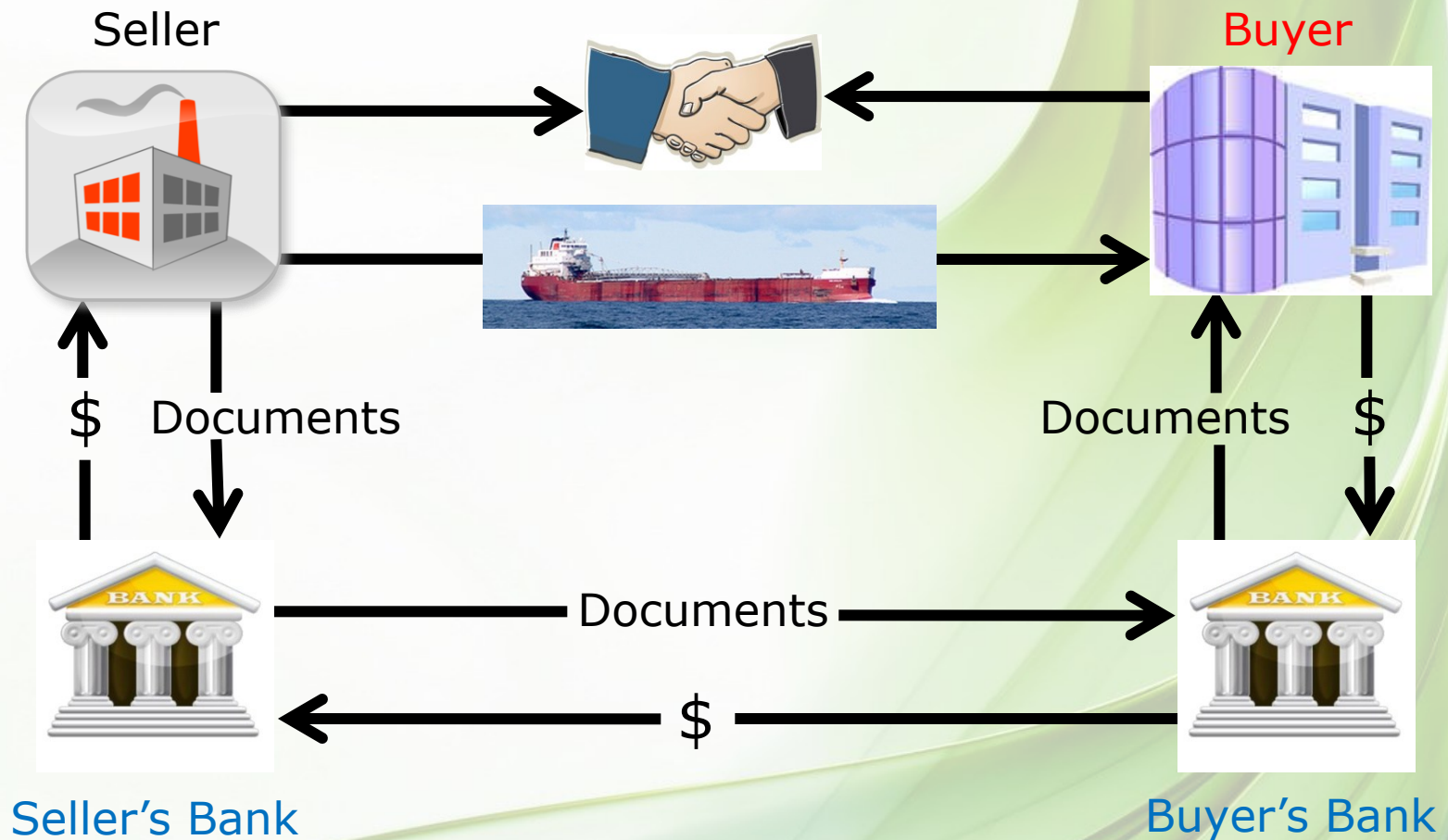
How about actual payments in
international trade?

Documentary Collection (D/C)

Characteristics

- Allows exporters to retain ownership of the goods until they receive payment or are reasonably certain they will receive it
- Bank acts as agent for exporter by holding the title documents
- Bank assumes no risk but must act in good faith and exercise reasonable care

Documentary Collection



Simplified D/C Transaction Flow

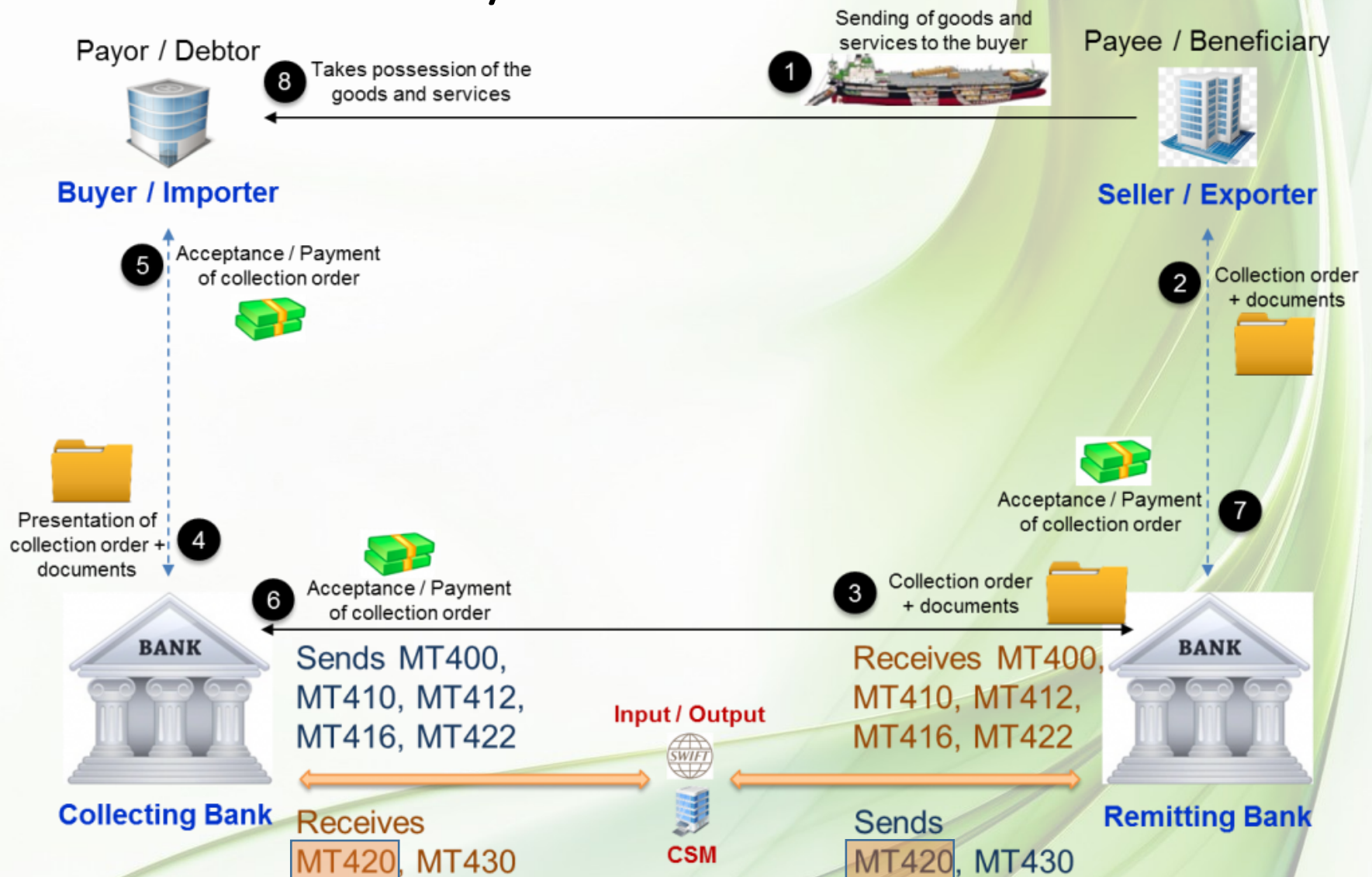
- The exporter ships the goods to the importer and receives the documents in exchange.
- The exporter presents the documents with instructions for obtaining payment to his bank.
- The exporter's remitting bank sends the documents to the importer's collecting bank.
- The collecting bank releases the documents to the importer on receipt of payment or acceptance of the draft.
- The importer uses the documents to obtain the goods and to clear them at customs.
- Once the collecting bank receives payment, it forwards the proceeds to the remitting bank.
- The remitting bank then credits the exporter's account.

Documentary Collection (2)

- Basic types:
 - Documents against Payment (D/P)
 - Documents against Acceptance (D/A)

Example: A Company from China exports goods or services to a French company

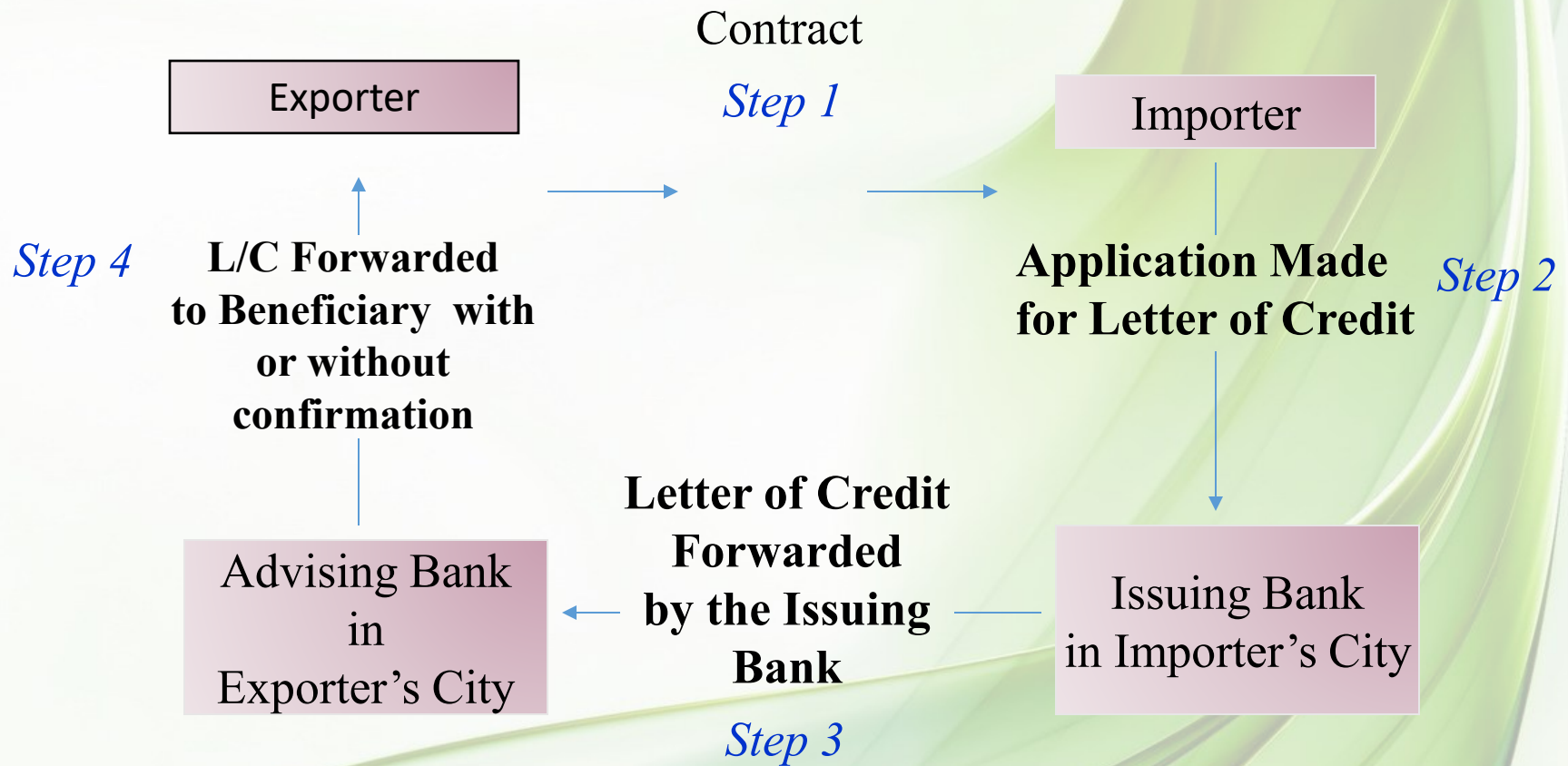
SWIFT & Documentary Collections



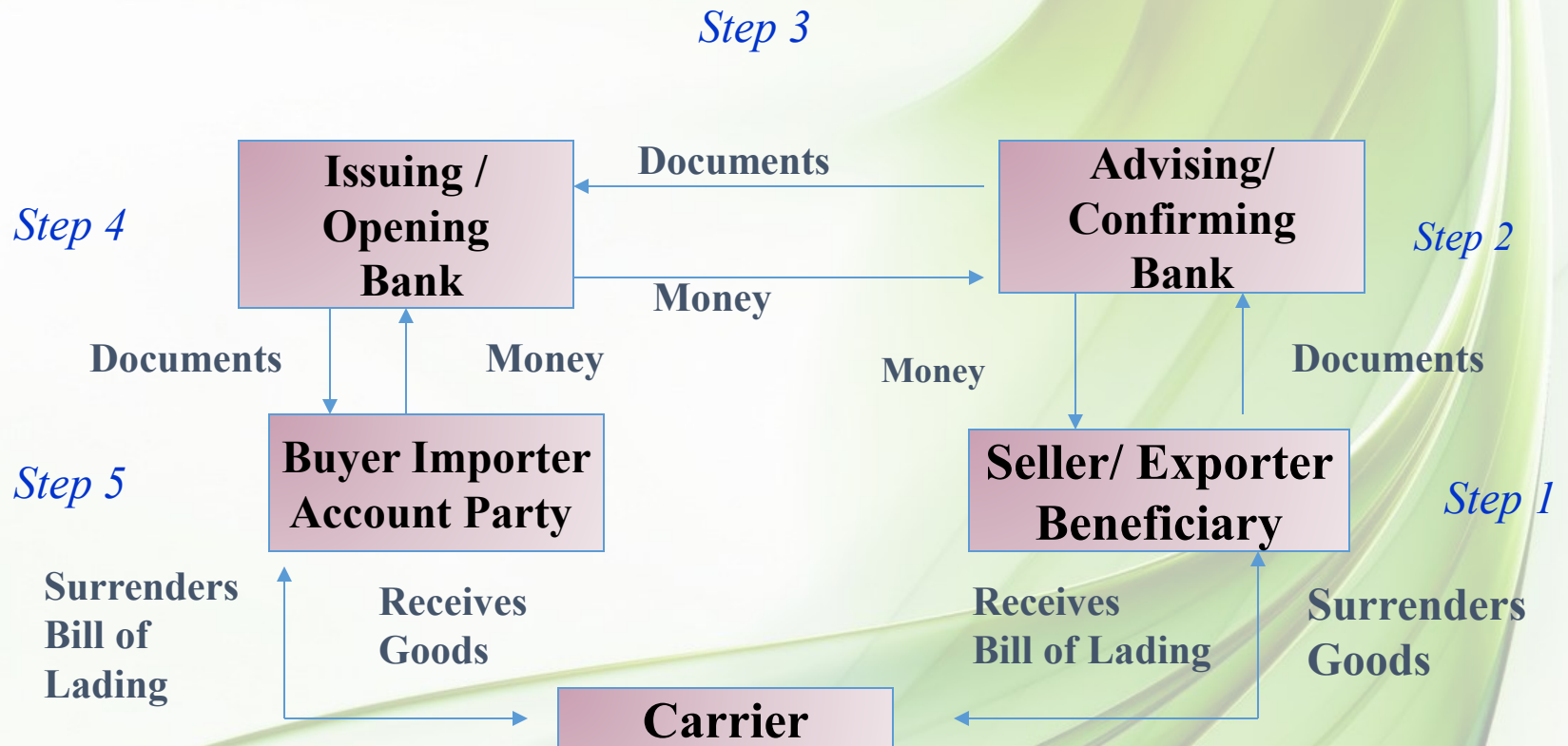
Letter of Credit

- **Letters of Credit (L/Cs)** are **legal instruments** issued **by banks (on behalf of their customers)** with the conditional obligation to make payment to the beneficiary of the L/C
 - **Documentary** (Trade) Letters of Credit are used to facilitate payments in import and export transactions
 - **Standby** Letters of Credit are “Standing By” for an event of default or non-performance before they can be drawn on

Letter of Credit Issuance



Letter of Credit Payment



Letter of Credit – Key Parties

- Applicant
 - Buyer/Importer
- Beneficiary
 - Seller/Exporter
- Issuing Bank
 - Guarantees payment
- Advising Bank
 - authenticates LC
- Confirming Bank
 - guarantees payment if Issuing Bank defaults

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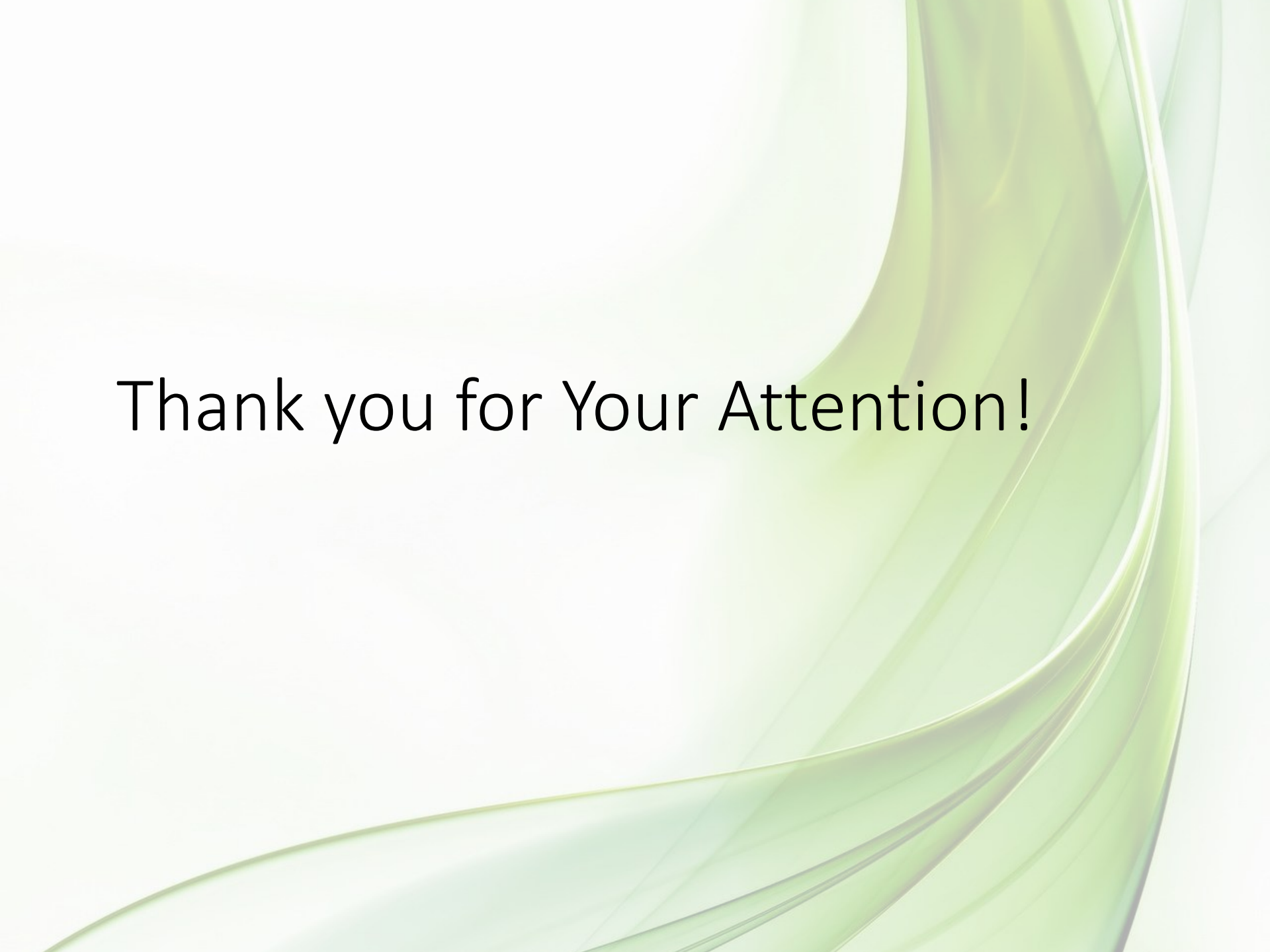
Trade Finance: Factoring/Forfaiting

Factoring/Forfaiting

- Methods used by businesses to convert outstanding receivables (invoices) into immediate cash, improving liquidity.
- Factoring:
 - Short-term, ongoing, often with recourse (seller bears risk)
- Forfaiting:
 - Long-term, transaction-specific, non-recourse arrangement
 - Used for high-value capital goods

Forfaiting

- Forfaiting is an export financing method where an exporter sells medium- to long-term receivables (usually 6 months to several years) to a financier (the forfaiter) at a discount, without recourse.
- **Key Features:** Non-recourse (forfaiter assumes all risk of default), 100% financing, often requires a bank guarantee.
- **Best For:** Large, one-time transactions, specifically capital
- **Example:**
 - A machinery manufacturer in Germany sells heavy equipment to a buyer in Brazil for €1 million, with payment due in 2 years.
 - The manufacturer cannot wait 2 years and needs cash now. They sell the payment obligation to a Forfaiter.
 - The Forfaiter immediately pays the manufacturer (e.g., €900,000, taking a discount) and takes on the risk.
 - When the Brazilian buyer pays in 2 years, the money goes to the Forfaiter

The background of the slide features abstract, flowing, organic shapes in various shades of green and white. These shapes curve and sweep across the frame, creating a sense of movement and depth. The colors range from light, airy whites to soft, muted greens, with some areas appearing more saturated than others. The overall effect is clean, modern, and visually appealing.

Thank you for Your Attention!

References

- Neal (2015): Concise History of International Finance: Chapters 2 & 3
- SWIFT website: <https://www.swift.com/>
- CIPS website:
<https://www.cips.com.cn/en/index/index.html>